



Digital twins for breast cancer treatment – an empirical study on stakeholders' perspectives on potentials and challenges

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Abstract

Purpose With 2.3 million diagnoses and 685,000 deaths annually, breast cancer is the most common cancer in women. The provision of necessary information throughout the whole patient journey is key to minimize the risk of breast cancer, to detect breast cancer as early as possible, and to aid the treatment process. Digital solutions provide abilities to holistically collect, transfer, and sophisticatedly analyze information. Specifically, digital twins in healthcare, as dynamic replicas of human bodies, are promising approaches for monitoring the condition of their patients and predicting tumor developments based on biometric data. However, the acceptance and adoption of such digital twin solutions in healthcare heavily depend on the individual stakeholders of the treatment process. This study aims to identify potentials and challenges of the introduction of digital twins in breast cancer applications from the involved stakeholders' perspectives.

Methods We conducted semi-structured interviews with 14 relevant stakeholders from the breast cancer treatment process. The interviews were then analyzed, based on the qualitative content analysis according to Mayring.

Results The results show that stakeholders see great potential in digital twin solutions to further facilitate personalized medicine, efficiency increases, and scientific benefits. However, the sensitive nature of healthcare causes numerous potential challenges in the technical, regulatory, user interface, and the strategic domain.

Conclusions The stakeholders unanimously agreed on the potential benefits of digital twins. However, existing systemic and individual stakeholder-level barriers hamper their introduction in breast cancer settings.

Keywords Digital Twin · Digital Health · Healthcare 4.0 · Breast Cancer · Personalized medicine · Empirical study

1 Introduction

Breast cancer is the most common cancer in women, with 2.3 million new cases and 685,000 deaths in 2020, and the leading cause of female cancer deaths in 107 of 183 countries (58%) [1]. Effective cancer control programs are rooted in the two approaches early diagnosis (to recognize symptomatic cancer) and screening (to identify asymptomatic diseases in apparently healthy target populations) [2, 3]. In addition, measures regarding prevention, treatment, palliative care, and survivorship care complement these approaches to a comprehensive cancer control plan

[3], needed to facilitate rapid and effective care [4]. Digital solutions play a central role throughout the patient journey. Enabled by the continuing rapid availability of medical data [5], new and emerging technologies to collect and process large amounts of data allow for unparalleled possibilities for personalized treatments [6]. A key technology leveraging medical data are digital twins. A digital twin presents a “living model of the physical asset or system, which continually adapts to operational changes based on the collected online data and information, and can forecast the future of the corresponding physical counterpart” [7]. They are predominantly used to simulate the behavior of physical objects in a virtual space, allowing for planning or optimizing without having to rely on the physical object [8].

In healthcare, digital twins allow for a dynamic (i.e., constantly updated) replica of the human body, or at least some parts of the human body [9]. With the development of appropriate simulation models and the availability of suitable data sets, the behavior of the human body and its subsystems can

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be simulated in order to not only monitor, but also predict its future state [10]. Digital twins will play a vital role in precision medicine and are already adopted in numerous experimental cancer care settings [11–13]. Because breast cancer diagnoses require the consideration of a multitude of risk factors [13], digital twins can play a major role in these treatments. Based on the integration of heterogeneous data sets into digital replicas, advanced analytics and artificial intelligence algorithms can provide (near) real-time scenario simulations along the patient journey (see Fig. 1).

Recent publications selectively pointed to various potential barriers in the introduction of digital twins in healthcare like policy, technical, and social aspects [14, 15]. While the technical development of digital twins for cancer care requires the collaboration between healthcare and technology experts [13], research has also shown that the consideration of a wide range of stakeholder perspectives during this development can provide valuable insights into potential real-world impacts [16]. Not only can the incorporation of various perspectives more accurately assess the potential

benefits and risks of a given technology [16], it also allows for tailoring the digital solution to the specific needs and preferences of the stakeholders [17]. Considering the sensitive nature of the healthcare system and its contribution to broader societal goals, such as health equity [18] or reducing healthcare costs [19], it is critical to holistically assess factors that might influence the introduction of a digital twin. Conclusively, this study aims to holistically identify potentials and challenges of digital twins in breast cancer applications along the patient journey from the perspectives of healthcare and technology experts. We therefore propose the following research questions: (I) What are the potentials of digital twins for breast cancer treatments? (II) What are the challenges of introducing a digital twin in breast cancer treatments?

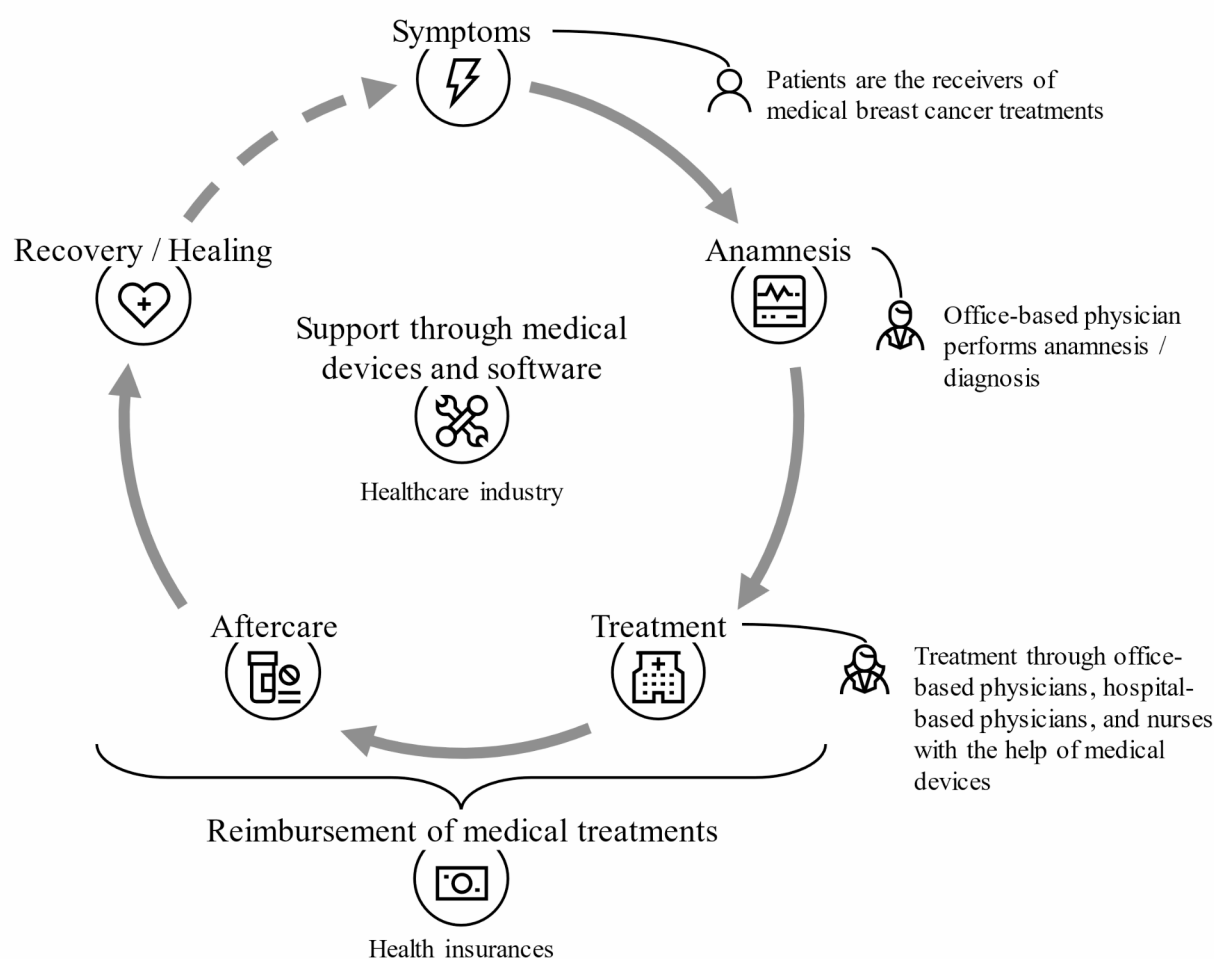


Fig. 1 Simplified patient journey in breast cancer care, highlighting essential stakeholders in the process

2 Methods

We conducted 14 in-depth interviews, ranging from 26 to 90 min (54 min average). The sample of interview partners (see Table 1) was composed to provide a holistic perspective of the key stakeholders along the patient journey of breast cancer patients in Germany (see Fig. 1). The interviews were conducted based on semi-structured guidelines with common questions for all participants and specific questions, tailored to the profession of the individual interview partner. Across all interviews, the general structure of the interviews was as follows. First, the interview partners were asked about their specific work environment. Second, they were asked about the current and prospective impact of digitalization on their work environment and specifically about digital twins in breast cancer applications. These parts of the interviews were tailored to the specific interview partners. E.g., the interview with a patient representative was focused on assessing the perspective of patients, the interviews with healthcare professionals focused on specific procedures in prevention, diagnosis, care, and aftercare, and the interview with the health insurance representative focused on data-related topics. In this part of the interview, the interviewer made the connection to the concept of digital twins and inquired about the possible integration of digital twins in the work environment, specifically asking about potentials and challenges.

Table 1 Composition of interview partners

Title, area of expertise (interview partner n = 14)	Interview code (see ESM ^a)
Hospital-based professionals (5)	
Chief physician, gynecologic oncology	IP06
Senior physician, obstetrics and gynecology	IP07
Specialist doctor, radiology	IP08
Specialist nurse, intensive care and anesthesia	IP09
Outpatient nursing service representative	IP10
Office-based physicians (4)	
Specialist doctor, internal medicine	IP02
Specialist doctor, internal medicine	IP03
Specialist doctor, gynecology	IP04
Specialist doctor, gynecology	IP05
Healthcare industry representatives (3)	
Product owner	IP11
Director, strategy and innovation	IP12
Senior director, strategy	IP13
Patient representative (1)	
Health law consultant	IP01
Health insurance representative (1)	
Innovation consultant	IP14

^aFor the full interview quotes of the individual interview partner, please refer to the Electronic Supplementary Material (ESM)

Based on a specific formal structure, the interview partner was first given the opportunity to speak freely by means of free and easy-to-answer questions. This was followed by more detailed inquiries about aspects that had already been discussed or new aspects that were relevant to the research topic. Finally, questions that were still open or not yet sufficiently discussed were addressed selectively. To keep the interview open and informative, the guideline served as a structure for the interview.

All interviews were transcribed and anonymized. For evaluation, qualitative content analysis according to Mayring [20] was chosen. The main categories of analysis (potentials of digital twins and challenges of digital twins) were derived from the research questions. Relevant statements in the interviews were assigned codes and initially segmented. Additionally, the identified statements were analyzed to form definitions and coding rules and finally collected into a coding agenda. More rigorous segmentation was used initially, categories were revised and summarized during the coding process to form the final structure.

3 Results

3.1 Potentials of digital twins in cancer care

3.1.1 Personalized medicine

In general, the interview partners agree on the potential benefit of a more individualized therapy by providing transparent risk and disease progression. One hospital-based physician highlights that by ascertaining a patient's personal risk factors, improved preventive measures could be introduced. This could prevent many diseases [ESM q#01],¹ especially in the context of an aging society [ESM q#02]. In addition, a digital twin could highlight own risk factors, which many patients often are not aware of [ESM q#03]. The anticipated advancements in prevention are expected to correlate with cost savings for the health system [ESM q#04].

3.1.2 Efficiency increase

Healthcare professionals and patient representatives alike see efficiency increase as potential benefits of digital twin-enabled cross-domain information management. For one, it would relieve patients of the burden to collect necessary information themselves [ESM q#05, ESM q#06], alleviating the chance of submitting incomplete or incorrect

¹ Please refer to the ESM for the full quotes of the respective interview partners. The interview codes from Table 1 match the title and area of expertise of the interview partners with the quotes in the ESM.

information to physicians [ESM q#07]. Others view the availability and timely access to aggregated and pre-analyzed information as major potentials [ESM q#07, ESM q#08]. Algorithm-based solutions could be used to assist the decision-making process of physicians [ESM q#09].

Having a near-complete picture of the patient would be particularly helpful for new patient admissions [ESM q#05] and could prevent costly and time-consuming duplicate examinations [ESM q#10]. In this context, a health insurance representative highlights the possibility of improved patient-physician relationships via a digital twin. This would also improve aftercare and could lead to cost savings, if the re-admission of patients to the hospital can be avoided [ESM q#11].

3.1.3 Scientific benefits

The interview partners also mentioned potential benefits to the scientific community. Healthcare professionals see value in using the collected data for research purposes [ESM q#12]. According to industry experts, collecting large amounts of data could allow for insights to be gained in a much better way and with a much higher probability of success than collecting small amounts of data [ESM q#13]. The involvement of a heterogenous set of partners would facilitate the generation of diverse data [ESM q#13]. Conclusively, the combination of large datasets from multiple sources as well as the inclusion of the latest findings from medical research could potentially feed back into much more accurate decisions and more individualized care for the patient [ESM q#14]. See Table 2 for a summary of the potentials of digital twins.

3.2 Challenges of digital twins in cancer care

3.2.1 Technical environment

Basic technical structure While the German telematics infrastructure provides the basic framework of the digital

infrastructure in the German healthcare system [ESM q#15], there is still a lack of suitable software interfaces that would enable the necessary exchange of data [ESM q#16]. Future digital twin solutions must be integrated into existing solutions [ESM q#17], even though these are still very cumbersome and hardly used at present [ESM q#18].

Data generation, data structuring, and data processing Physicians argue that the increase in data availability should not result in additional data handling tasks [ESM q#19], as the purpose of collecting data through a digital twin is to make preparation for a patient more efficient. Respondents believe structuring and evaluating all the data would put additional stress on staff [ESM q#20, ESM q#21]. Accordingly, minimum standards for data generation, collection, and storing should be developed. These standards should specify what data is necessary to generate a meaningful digital twin [ESM q#22, ESM q#23].

In addition, it is necessary to define standards for the data-collecting equipment [ESM q#24]. While mobile health technologies could help gather enough data on a person to create the basis for a holistic digital twin, a representative of the healthcare industry says that with this large amount of data comes the challenge of bringing the data together “in a kind of an organized way”, to “interpret, understand, and act on that data” [ESM q#25]. Advanced analytics are needed to prevent information overload [ESM q#26]. These challenges are accompanied by potential increases in direct costs for the devices and digital services [ESM q#27].

Data consolidation The heterogenous nature of healthcare data raises the question of central access for relevant stakeholders. While healthcare professionals favor a centralized storage location under the sovereignty of the patient [ESM q#28, ESM q#29], the specific technical architecture needs to be discussed. Suggestions ranged from cloud-based solutions [ESM q#28] to local storage on the health insurance card [ESM q#30].

Interoperability While respondents already raised the question of central data access, the interoperability, e.g., the ability of all involved systems to work together seamlessly, is also seen as a major challenge [ESM q#31, ESM q#32, ESM q#33]. This also includes ensuring ease of use on the patient side [ESM q#34], as well as at partners outside the hospital environment, involved in the treatment process [ESM q#35]. A representative of a health insurance company emphasizes that it is a challenge to enable a mandatory technical connection of all service providers [ESM q#36].

Table 2 Dimensions of potentials of digital twins in breast cancer care

Dimensions of potentials	Examples
Personalized medicine	- Improved care through the ability to collect large amounts of data across entities - Improved prevention by highlighting risks and possible disease progression
Efficiency increases	- Increased efficiency in data collection - Increased availability of necessary data - Reduced need for readmissions by improved communication and better follow-up-care
Scientific benefits	- Enabled large scale analyses of data - Generation of diverse datasets - Feed back of analyses into improved personalized care

3.2.2 Regulatory integration

Data privacy Health data are sensitive. To prevent negative effects for patients, it is important to regulate the rights to access and change the data [ESM q#37, ESM q#38]. This has already led to increasingly difficult and complicated data protection regulations [ESM q#39]. It also leads to the emergence of questions regarding data ownership [ESM q#40]. While many respondents are in favor of data sovereignty resting with the patient [ESM q#41, ESM q#42], it can be assumed that physicians fear to only receive incomplete information [ESM q#43]. Patients might be unable to identify relevant documents for the physician and block these documents from the physician or even delete relevant data [ESM q#44]. Widespread data autonomy would require the patient to have the competence to decide which actor must receive which data [ESM q#45].

Questions regarding the storing of health data and how the intersectoral exchange of data has to take place must also be addressed. Respondents see ethical challenges emerging, e.g., as soon as health insurance companies are able to acquire even more diverse datasets, a kind of transparent patient could emerge [ESM q#37]. Analytical models based on digital twin data could result in health insurers' ratings of individual patients, which must be avoided [ESM q#46]. Respondents see themselves caught in a quandary of patients' autonomy over their own data [ESM q#47], and a simple, practicable, and manageable solution.

Billing of digital services With digital technologies receiving little support in the German healthcare system [ESM q#48], an office-based physician urges the need for legislation to reward the efforts involved in handling these technologies [ESM q#49]. A nurse also believes that the scope of services should be paid based on performance [ESM q#50]. One office-based physician believes that the healthcare system has changed a great deal in recent years. As a result, costs continue to rise, which she can see in the ever-increasing bills [ESM q#51]. However, digital issues are currently given little consideration in the healthcare financing system [ESM q#52].

Financing the digital twin During the interviews, the issue of financing digital technologies in the healthcare sector became apparent [ESM q#53]. One interview partner asked how a digital twin will be financed and refinanced [ESM q#54]. Hospitals so far have little budget to pay for the development [ESM q#55]. A nurse said that while purchasing new systems is often expensive, it would save money in the long run because outdated systems are often inefficient and require a significant amount of time [ESM q#56]. A representative of the healthcare industry is of the opinion that,

at the current time, the health insurers should be the payers of a digital twin. However, according to the current state of the incentive system, the investment system, and the financing system, this does not seem realistic to him [ESM q#57]. Also, it would not be in the economic interest of a health insurance company to make upfront investments that lie so far in the future [ESM q#58]. A different representative of the healthcare industry proposed that the industry should try to develop business models that could generate sustainable revenues. Thus, most of the development costs would have to be paid by the company, unless they enter partnerships through which they are able to offset the costs [ESM q#59]. It is important to initiate a conversation with various stakeholders and interest groups and to demonstrate the value of the concept [ESM q#60].

3.2.3 User interface

Interview partners raised concerns regarding the human-machine interaction of digital twins, like the user-friendliness of interfaces. A patient representative cites the reservations about new digital technologies on the physicians' side. Physicians might fear high and time-consuming investments due to the complexity of concepts like a digital twin [ESM q#61]. A physician agrees and states that a solution should support the treatment process and provide enhanced information capability to spend more time with the patient [ESM q#62].

User interface also applies to the patient side. According to a patient representative, it is a balancing act between user-friendliness and data protection. While data sovereignty on the patient side would comply with data protection regulations, too much autonomy could reduce user acceptance, as they might not want to deal with these issues in such depth [ESM q#63]. A secure, yet simple manageable solution is needed [ESM q#64, ESM q#65]. Interview partners proposed an app-based solution [ESM q#66] with a user interface based on the needs of the users, as a viable solution [ESM q#67]. It is important to take patients by the hand through all steps of care, and preferably all with one platform [ESM q#68], complemented by information or training videos for new tools [ESM q#69].

Additionally, an office-based physician argues that it would be important to demonstrate the added value that the use of a digital twin brings with it to all stakeholders [ESM q#70]. Physicians from the hospital also believe that benefits, such as financial incentives, could help motivate patients to use a digital twin [ESM q#71]. The future incentivization of patients to participate in a digital twin setting and check-ups, which could be a source of data for a digital twin, remains unclear [ESM q#72].

3.2.4 Complementary objectives of stakeholders

For a digital twin to succeed, it is necessary that all relevant stakeholders are involved in the development process and share a common goal. Specifically, an office-based physician states that it is important that all players pull together to implement a digital twin and commit to the necessary networking of all players [ESM q#73]. A representative of the healthcare industry says that a holistic approach can only work if policymakers and industry work together to promote a holistic and centralized system [ESM q#74]. Another interview partner emphasizes that it will be a great challenge to unite the interests of different stakeholders [ESM q#75].

However, because of the high number of relevant stakeholders in the healthcare system, it is challenging to unite each stakeholder's vision for digital care. Not only remains the question of ownership of the digital twin solution, but also how to coordinate all relevant stakeholders [ESM q#76, ESM q#77]. See Table 3 for a summary of the challenges of digital twins.

4 Discussion

The lack of existing comparable solutions to the digital twin in breast cancer applications is forcing the interview partners to evaluate potentials and challenges from a more abstract point of view. While these results can be extrapolated to other medical applications of digital twins, this does not imply that the relevance for breast cancer application is diminished.

Referring to research question (I), specific potentials of digital twins were aggregated into three main dimensions: personalized medicine, efficiency increasements, and scientific benefits. Personalized medicine based on digital twin

technologies is still in its infancy and perceived as a trial-and-error approach [21]. With a void of relevant literature on digital twins in real-world (breast) cancer applications, the magnitude of potential ethical issues is hard to assess. Current digital solutions are vulnerable to discriminatory outcomes or distorted views on health [22].

The same applies to the anticipated increases in efficiency. While the identified aspects are also reflected in assessments of current literature (e.g. [23]), juristically approved digital twin-based medical solutions are scarce [24], especially in breast cancer settings. Real-world evidence on desired benefits for individuals or the system itself has yet to emerge.

The potential of advancing the scientific field, an abstract socio-ethical benefit, is traditionally given less attention in scientific discussions as it constitutes only as an adjacent benefit for the involved stakeholders than a goal itself [14]. Still, a sophisticated technical infrastructure would be needed. As stated by interview partners, the questions of where the data is located and how the intersectoral exchange of data has to take place must be answered. Despite the growing interest from researchers and practitioners, academic discussions are still void of a standardized definition and a reference architecture [25]. Promising architectures like data spaces and federated data ecosystems could provide the necessary technological underpinnings for digital twins in healthcare. Yet these architectures are still to be established in healthcare and the necessary collective data governance is particularly challenging to introduce [26, 27].

Answering research question (II), identified challenges of digital twins in breast cancer applications were aggregated into four dimensions: technical environment, regulatory integration, user interface, and complementary objectives of stakeholders. With respect to the regulatory integration and the financing and billing of digital services, the results suggest that even though the financial aspects like clear regulations regarding billable services of physicians are important for the successful implementation of a digital twin, regulators in Germany have to implement further opportunities to charge for digital services and products. This lack of reimbursement policies is not unique to the German healthcare system, as, e.g., the USA also have been slow to adapt to technological advancements in digital health solutions [28]. Rigorous restrictions to the coverage of digital services from both, public and private health insurers, are a characteristic of many western healthcare systems and hamper the introduction of such technologies. Given the continuous advancements of healthcare technologies, the need to explicitly consider the impact on health equity emerges [29]. Because health outcomes are widely effected by social determinants such as income and education [30],

Table 3 Dimensions of challenges of digital twins in breast cancer care

Dimensions of challenges	Examples
Technical environment	• Guarantee compliance with existing solutions • Reduce burden of data generation, data structuring, and data processing • Data consolidation governance • Interoperability of involved systems
Regulatory integration	• Ensure data privacy and autonomy • Enable the value capture of digital solutions • Financing of core digital twin services
User Interface	• Consideration of users' preferences during development • Accessibility of digital solutions • Uncomplicated, tangible and easy-to-handle-solution
Complementary objectives of stakeholders	• Ensure commitment of all relevant stakeholders • Coordination of collaborative efforts

new solutions should always seek financing strategies to reduce health disparities.

The interview partners also raised concerns regarding ethical aspects that go beyond financial considerations. In line with recent academic discussions on ethics of big data in health care [31], interview partners highlighted the need for appropriate safeguards and regulations to prevent discriminatory outcomes. Regulators must ensure a proper balance between respecting patients' autonomy and rights and allowing the exploitation of patients' individual data to contribute to a broader understanding of health care to benefit society as a whole [32].

5 Limitations

This study is not without limitations. The sample was slightly dominated by medical experts, with industry, patient representative, and insurance experts accounting to approximately a third of all interview partners. Based on the quadruple helix theory [33], this dominant helix is typically motivated to prioritize medical outcomes and to further academic knowledge, rather than economic value for individual actors in the healthcare system. However, the results originate from valid perspectives of relevant actors within the patient journey. At least *prima facie*, the responses of the interview partners were not restricted to their helix of origin. This is reflected in the diversity of addressed topics. While outside the scope of the present study, future investigations could provide an even broader perspective by including representatives from, e.g., the pharmaceutical industry, regulatory bodies, or patient support groups. Also, as only experts from Germany were included into the sample, the generalizability of the results might be limited to western and developed countries, as healthcare systems from low- and middle-income countries differ significantly from high-income systems [32].

6 Conclusion

This study identifies stakeholders' perspectives on potentials and challenges of the introduction of digital twins in breast cancer applications. Three dimensions of potentials (personalized medicine, efficiency increasements, and scientific benefits) and four dimensions of challenges (technical environment, regulatory integration, user interface, and complementary objectives of stakeholders) could be identified. The interview partners unanimously agree on the prospective beneficial nature of a potential twin in breast cancer applications and for the medical field in general, but also call for the necessary requirements.

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Declarations

Competing interests Larissa Wolf was employed at a medical device manufacturer while the study was conducted. Jens Konopik and Oliver Schöffski have no conflict of interest.

Ethical approval Non applicable.

Consent to participate Informed consent was obtained from all individual interview partners included in the study.

Consent to publish The authors affirm that all interview partners provided informed consent for publication.

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